Math 110 Fall 2026 Running Review Document

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August 31, 2026

1 Disclaimer

This review document will be based on the lectures of Alan Hammond in the Fall of 2026 (our course). These lectures are themselves based on the lectures of Edward Frenkel, and Sheldon Axler's *Linear Algebra Done Right*. These are supplemetary and are not a replacement for the lecture notes, or for attending lecture. They are meant as a short recap with some guiding questions based on my personal understanding of the content.

2 Sets and Proofs

A set, in its essence, is just some collection of objects. That is the definition we use in discrete math, and for many purposes, though it is a naive definition (Renee Descarte and several others brought up paradoxes in this basic idea, leading the the creation of the Zermelo-Fraenkel axioms as the foundation of set theory). Most of the set theory we need is a review of Math 55, Discrete mathematics. You should be comfortable using set notation, like saying $x \in M$ for "x is an element of M", or using set-builder notation like $\{(x, x^2) \mid x \in \mathbb{R}\} \subset \mathbb{R}^2$ (*what is this set?*). You should also recognizes unions ($A \cup B$), intersections ($A \cap B$), complements (A^C), set differences ($A \backslash B$), and cartesian products ($A \times B$). This is also a good time to review the basics of proofs and generally interesting spaces, like the natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$, and the real numbers $\mathbb{R}$.

- 1.) Prove that $\left(\bigcup_{a\in A} U_a\right)^C = \bigcap_{a\in A}(U_a^C)$. (This is DeMorgan's law)
- 2.) Define injective, surjective, and bijective. Construct a function which is injective but not surjective. Construct another that is surjective but not injective.
- 3.) Construct a function $f : \mathbb{N} \to \mathbb{Z}$ to show that these two sets have the same cardinality (i.e. construct a bijective function).

Throughout this course, we will be using proofs. Proofs are a formal argument that gives evidence for some mathematical statement. The following is an example of such a proof presented during lecture.

Example: x^2 is even if and only if x is even.

Proof: Assume first that x is even. Then, $x = 2k$ for some integer k, and $x^2 = (2k)^2 = 4(k^2) = 2(2k^2)$. Since $m = 2k^2$ is also an integer, $x^2 = 2m$ is even. This proves the statement "If x is even, then x^2 is even".

Since we claimed an "if and only if", we also need to prove the claim "if x^2 is even, then x is even". We do this by proving the *contrapositive*, that if x is not even (odd), then x^2 is not even.

Indeed, if x is not even, $x = 2k+1$ for some integer k. Then, $x^2 = 4k^2+4k+1 = 2(2k^2+2k)+1 = 2m+1$ for $m = 2k^2 + 2k$ an integer. Therefore, x^2 also fails to be even.

The essence of our proof here is to expand on the definition in our assumption for each case, then use some other structure or calculation to show that some criteria follows from this definition. We will do this very often. One important method of proof for our course is called *proof by induction*. Mathematical induction (not electrical) is the concept that if we want to prove some condition R_n for every number $0, 1, 2, 3, 4, 5, ...$, we first show that R_0 holds (our "base case"), then show that if R_n holds, then R_{n+1} does ($R_n \Rightarrow R_{n+1}$). A great example is the following:

Example: For every nonnegative integer n,

$$\sum_{i=0}^{n} i = \frac{n(n+1)}{2}$$

First, when $n = 0$, the sum is empty and so clearly both sides are 0. Next, assume that this holds for n. Then,

$$\begin{aligned}\sum_{i=0}^{n+1} i = (n+1) + \sum_{i=0}^{n} i &= (n+1) + \frac{n(n+1)}{2} \\ = \frac{2(n+1) + n(n+1)}{2} &= \frac{(n+1)(n+2)}{2}\end{aligned}$$

so that the condition holds for $n+1$. Therefore, by the principal of mathematical induction, we have the claimed statement.

You should practice writing proofs, as they are the main mode of communicating mathematical facts.

- 1.) Show that for all non-negative integers n,

$$\sum_{i=0}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

- 2.) Prove that $\mathbb{Z} \times \mathbb{Z}$ has the same cardinality as $\mathbb{Z}$.
- 3.) Prove that there are infinitely many prime numbers. *Hint:* Assume that there are finitely many primes $p_1, ..., p_n$. Can you find a prime not in this list? If so, you have found a *contradiction* and your assumption must be false.

3 Fields

Fields are a special type of set with very nice addition and multiplication. In particular, they have structure built to generalize the properties of the real numbers. Fields are a *unital ring* in which every nonzero element has a multiplicative inverse. Let's break down this definition. The "ring" part means that addition and multiplication are defined on the elements of the ring (where addition is commutative and associative, and multiplication is associative and distributive). This addition also has an additive identity 0 so $o + r = r$ for a ring element r, as well as an additive inverse $-r$ so $r + (-r) = 0$. Next, "unital" means we have a multiplicative identity 1 so $1(r) = r$ for every element r of our field F, and the multiplicative inverse means for each r that is not zero, we have $\frac{1}{r}$ so $r\frac{1}{r} = 1$. These fields are very important in abstract algebra, so it is worth looking over the full axiomatic definition a few times to understand it. We will mostly be working over the fields $\mathbb{R}$ and $\mathbb{C}$ (the complex numbers), though we will retain some abstraction. The complex numbers are particularly important because they are what is called an *algebraically closed* field, which just means that every polynomial with complex numbers as coefficients has all of its roots in the complex numbers. This fact is called the Fundamental Theorem of Algebra (a proof of it often relies on a set of theory called *complex analysis*, which is a bit beyond this course). You should recall how to multiply, add, conjugate, and divide by complex numbers.

- Which of the following are fields? $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R} \times \mathbb{R}$ (is multiplication defined here?), $\{0, 1\}$
- Consider $\mathbb{Z}_3$, which is the set $\{0, 1, 2\}$ where addition and multiplication are define modulo 3 (so $1 + 2 = 0$, $2 + 2 = 1$, $1 + 1 = 2$). Is this a field?
- Let $a = 1 + 3i$ and $b = 4 + 5i$. Compute

$$\overline{a}\frac{1}{b}$$

- Show

$$\left(\frac{-1 + i\sqrt{3}}{2}\right)^3 = 1$$

- Show $0a = 0$ for all $a \in \mathbb{F}$. *Hint:* $0 = 0 + 0$
- Show that if $1 \in \mathbb{F}$ has additive inverse -1, then $(-1)a = -a$ for all $a \in \mathbb{F}$.

The "standard" structure we will use from this point on and compare many things to is $\mathbb{F}^n$, where $\mathbb{F}$ is a field. This is the set of n-tuples or lists $\{(a_1, ..., a_n) \mid a_i \in \mathbb{F}\}$. Addition is defined component-wise, so

$$(a_1, ..., a_n) + (b_1, ..., b_n) = (a_1 + b_1, ..., a_n + b_n)$$

and we define a *scalar multiplication* by elements of the field: for $\alpha \in \mathbb{F}$,

$$\alpha(a_1, ..., a_n) = (\alpha a_a, ..., \alpha a_n)$$

- 1.) What is $\mathbb{R}^3$?
- 2.) Show that this multiplication is distributive, i.e. $\lambda(a + b) = \lambda a + \lambda b$ for $\lambda \in \mathbb{F}$, and $a, b \in \mathbb{F}^n$.
- 3.) Show $1x = x$ for $x \in \mathbb{F}^n$.
- 4.) For $x \in \mathbb{F}$, show that $x^n = x(x)(x)...(x) = 0$ for some n if and only if $x = 0$.

Remark: The reason we bother with this abstract definition of a field is that there are many other fields, and they can provide cases that work against our intuition. For example, we have the integers modulo a prime. "modulo" is an operation that means "the remainder when dividing by". If we are working modulo 5, we have

$$\begin{aligned}
14 + 27 = 41 = 1 + 6(5) &\equiv 1 \mod 5\\
2 * 4 = 8 = 3 + 1(5) &\equiv 3 \mod 5\\
5 = 0 + 1(5) &\equiv 0 \mod 5\\
4 &\equiv 4 \mod 4\\
4 = (-1) + 1(5) &\equiv -1 \mod 5
\end{aligned}$$

The last three cases show that working modulo 5 only requires the numbers $\{0, 1, 2, 3, 4\}$. We call the integers mod 5 by the name $\mathbb{Z}_5$. This is also a field (check the axioms, e.g. $2 * 3 \equiv 1 \mod 5$ gives a multiplicative inverse for 2) that is important in a field of math called abstract algebra.

4 Vector Spaces

4.1 Definition

A vector space is a broader abstract concept of a space with "additive structure", but the multiplicative structure is through *scalar* multiplication and so the space isn't quite as nice as just a ring. Our prototypical vector space is the $\mathbb{F}^n$ we defined above.

For any vector space, we begin with some set V and define two operations on the set. The first is addition $+ : V \times V \to V$, which is defined to be commutative and associative, have an additive identity element 0, and have for each $v \in V$ an inverse $-v$ so $v + (-v) = 0$. Next, any vector space is associated with a field $\mathbb{F}$, and we call V a vector space over the field $\mathbb{F}$. This gives the second operation, scalar multiplication $* : \mathbb{F} \times V \to V$ which is defined to be associative and distributive, while we also assume $1 * v = v$ for every vector $v \in V$.

You should note that the additive identity 0 is unique, so there cannot be two distinct vectors with the additive identity property.

In the case of $\mathbb{F}^n$, the addition was component-wise addition of vectors and the multiplication was component-wise multiplication by the field element. Therefore, all of these properties follow from the field axioms. In the future, we will actually show that many vector spaces (the finite dimensional ones) look like (are isomorphic to) our example $\mathbb{F}^n$, which is why we keep bringing this example up.

- 1.) Using the vector space axioms, show that the set of 2×2 matrices whose elements are real numbers, $M_2(\mathbb{R})$, is a vector space.
- 2.) Is $\mathbb{R}$ a vector space over $\mathbb{C}$? Is $\mathbb{C}$ a vector space over $\mathbb{R}$?
- 3.) Like for fields, show $0v = 0$ for every $v \in V$ and $(-1)v = -v$ for all $v \in V$.

Vector spaces are the foundation of our understanding of a "linear space" where we want to study linear structure. This structure seems weird and abstract at first because it is- it took many mathematicians working on affine geometry before such definitions were made. While Rene Descarte and Pierre de Fermat began identifying system solutions with plane curves as early as 1636, removing coordinates from these processes (i.e. using abstract objects) first began with Bolzano in 1804, and the modern definition of the vector space was first adopted by Giuseppe Peano in 1888 based on the work of Cayley on matrices (1857) and Grassmann's abstraction of addition and scalar multiplication (1844).

4.2 Subspaces

If we consider a vector space like $\mathbb{R}^3$, we may also notice that things which look like other vector spaces are contained in $\mathbb{R}^3$. For example, $\{(0, y, z) \mid y, z \in \mathbb{R}\}$ looks very much like $\mathbb{R}^2$. This brings us to the concept of a ***subspace***. A subset U of a vector space V is a subspace if it is a vector space with the operations inherited from V. This means that, when checking to see if a subset is a subspace, we don't need to check whether the addition is associative or the multiplication is distributive, we just need to check that these operations are defined on U. In particular, we need to check that for all $u, v \in U$ and $a, b \in \mathbb{F}$, $au + bv \in U$.

- 1.) Prove that, given $U \subset V$ satisfies this criterion, that U is a vector space.
- 2.) Given a 2×2 matrix A, show that the solution set

  $$\{x \in \mathbb{R}^2 \mid Ax = 0\}$$

  is a subspace. Show that this fact is not true in general if we look at x solving $Ax = [1, 0]$.

4.3 Sums

This definition is quite broad and gives many more subspaces than the simple form above. From Calc 3, you may remember that all planes and lines in $\mathbb{R}^3$ are subspaces. Further, if we consider the space of all functions from $[0, 1] \to \mathbb{R}$ with pointwise addition and scalar multiplication, denoted $\mathbb{R}^{[0,1]}$, the continuous functions are a subspace and the differentiable functions are a subspace of the continuous functions (and so a subspace of the original vector space).

There are many other ways to produce subspaces. For example, the intersection of subspaces is also a subspace. This is not true in general for unions: consider $A = \{(x,y) \mid y = 0\} \cup \{(x,y) \mid x = 0\}$. This is the union of the x and y axis in $\mathbb{R}^2$. Each component of the union is a subspace, but $(1,0)+(0,1) \notin A$ shows that A is not closed under addition.

A better way to form a larger subspace is to instead take the sum of subspaces. Given $V_1, ..., V_m \subset V$ that are subspaces,

$$V_1 + ... + V_m = \{v_1 + ... + v_m \mid v_i \in V_i \text{ for all } i\}$$

In particular, the sum is the *smallest* subspace containing each of the constituent subspaces, which is to say that if $W \subset V$ is a subspace satisfying $V_i \subset W$ for all i, then $V_1 + ... + V_m \subset W$.

- 1.) What is $\{(x,y) \mid y = 0\} + \{(x,y) \mid x = 0\}$?
- 2.) Prove that the intersection of subspaces is still a subspace.
- 3.) Consider the real vector space $R = \{(a_0, a_1, a_2, ...) \mid a_i \in \mathbb{R},\ \sum |a_i| < \infty\}$. Show $R_i = \{(a_0, a_1, ...) \mid a_j = 0 \text{ for } j \neq i\}$ is a subspace. Find vectors $v_i \in R_i$ so $\sum_{i=1}^{\infty} v_i \notin R$, and so we cannot extend the above definition to infinite sums without some care.
- 4.) Prove that the union of two subspaces is a subspace if and only if one subspace contains the other.

Recall that we define $\mathbb{F}^n$ using *coordinates* $(a_1, ..., a_n)$. This is a very special case of a sum where $\mathbb{F}^n = \sum \mathbb{F}_i$ for

$$\mathbb{F}_i = \{(0, ..., 0, a_i, 0, ..., 0) \mid a_i \in \mathbb{F}\}$$

The special part is that these subspaces don't have a "large" overlap, which means that any two people looking at the same point will write the same coordinates with respect to this sum.

For example, if we add

$$\{(0,0,z,a) \mid z, a \in \mathbb{R}\} + \{(x,y,0,0) \mid x, y \in \mathbb{R}\}$$

we get all of $\mathbb{R}^4$, but we could just as well add

$$\{(x,y,0,0)\} + \{(0,y,z,a)\}$$

The way we can symbolically notice the overlap in the subspaces is that, in the first case, we may decompose any vector

$$(x,y,z,a) = (x,y,0,0) + (0,0,z,a)$$

as the sum of two vectors from the two subspaces, and there is only one possible way to do this. In the second case

$$(x,y,z,0) = (x,y-1,0,0) + (0,1,z,0) = (x,y-2,0,0) + (0,2,z,0) = ...$$

The overlapping parts of the vector space mean that we can take infinitely many different decompositions.

This leads us to the definition of a *direct sum*. Suppose $V_1, ..., V_m$ are subspaces of V. The sum $V_1 + ... + V_m$ is called a direct sum if each element $v \in V_1 + ... + V_m$ may be written uniquely as the sum $v = v_1 + ... + v_m$ for $v_i \in V_i$. In this case, we denote the sum $V_1 \oplus ... \oplus V_m$.

In particular, $U + W$ is direct if and only if $U \cap W = \{0\}$.

- 1.) If $V_1, V_2, W \subset V$ are subspaces such that $V = V_1 \oplus W = V_2 \oplus W$, is it true that $V_1 = V_2$ (try experimenting in $\mathbb{R}^2$).?
- 2.) Consider $\Lambda = f : \mathbb{R}^2 \to \mathbb{R}$. Let $A = \{f \mid f(x,y) = -f(y,x)\}$ and $S = \{f \mid f(x,y) = f(y,x)\}$. Show $\Lambda = A \oplus S$ (we will look at this example again much later).
- 3.) Consider $W = \{(x,y,x,y) \in \mathbb{R}^4\}$. Find a subspace V so $W \oplus V = \mathbb{R}^4$.

5 Linear Independence and Span

6 Dimension

7 Linear Maps

8 Matrices

9 Isomorphisms

10 Products, Direct Sums, Quotient Spaces